

१. What are the different sources of water? What are the considerations to be taken in selecting the source of water?

Answer:

- Any natural or artificial source from which water can be utilized for domestic, industrial, agricultural, recreation and other purposes is known as sources of water supply.
- Most of the water obtained through precipitation is retained in surface depressions, carried away as surface runoff in natural streams or rivers and percolates into the ground which joins groundwater.
- Hence, sources of water can be broadly classified into following types. They are:
 - i. Surface water source:
 - Water available on the ground surface is known as surface source.
 - The quality and quantity of water available on the surface source depends on the rainfall patterns, climatic and geological factors.
 - Various forms of surface sources of water are:
 - ✓ Lakes and ponds
 - ✓ Stream or rivers
 - ✓ Storage or impounded reservoirs
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 - ii. Groundwater source:
 - Water that exists below the ground surface is termed as groundwater source.
 - The quality of groundwater is generally good due to natural infiltration as it infiltrates deep into the soil percolating into deeper strata.
 - Groundwater acquires its chemical characteristics from surface water that percolates into the ground.
 - Various forms of the ground sources are as follows:
 - ✓ Springs
 - ✓ Infiltration galleries
 - ✓ Infiltration wells
 - ✓ Wells and tube wells

The major factors to be considered while selecting the sources of water supply are as follows:

- i. Quality of water:
 - The source should be able to supply a sufficient quantity of water to meet the various demands of the city during the entire design period.
 - Water availability in the source may vary with the season to season so the quantity of water available in dry season should be adequate to meet the water demand, i.e., safe demand.
- ii. Quantity of water:
 - The source should have safe wholesome, free from pollution of any kind and other undesirable impurities.
 - The impurities present in the water should be as less as possible and these should be removed easily and cheaply.
- iii. Location:
 - As far as possible source of water should be locating near to the community and it should be situated in an elevated area.
 - This will reduce the length of pipe and water from the source would flow by gravity and hence the cost of the project reduced.
- iv. Cost of water supply projects:

- It should be able to supply water of good quality and quantity at the less cost.
- The cost depends upon the location, distance between city and the source, distribution area etc.

2. **Write short notes on:**

a. Ph value

b. Water demand

c. Hydraulic grade line

Answer:

a. pH value:

pH is a measure of the acidity or alkalinity of a solution. It indicates the concentration of hydrogen ions (H^+) present in water.

$$pH = -\log_{10}[H^+]$$

where $[H^+]$ is the hydrogen ion concentration $\in$ moles per liter.

If,

pH < 7: Acidic solution

pH = 7: Neutral solution (pure water)

pH > 7: Alkaline (basic) solution

The pH scale ranges from 0 to 14.

Importance of pH:

- i. pH indicates the quality of drinking water.
- ii. It influences the effectiveness of water treatment and chlorination.
- iii. It also affects corrosion and scaling in pipelines.
- iv. pH is essential for aquatic life and irrigation water quality.
- v. It is important in chemical and biological reactions.

Measurement of pH can be done by using pH meter, Ph paper or universal indicator.

According to WHO guidelines, the desirable pH range for drinking water is 6.5-8.5.

Application of pH:

- i. pH is used in drinking water quality monitoring.
- ii. It is used in wastewater treatment
- iii. Environmental monitoring
- iv. Agriculture and irrigation
- v. Industrial process control

b. Water demand:

Water demand is the total volume of water required to fulfill various human activities and sustain ecological requirements. The daily water demand for Kathmandu Valley is more than 360 million liters per day, whereas the supply is around 90 million liters per day in dry season. It indicates that water demand has grown faster than reliable water supply.

Water demand is necessary for designing the water supply schemes. There are different types of water demand such as domestic demand, livestock demand, municipal demand etc. In Nepal, domestic demand in fully plumbed house is taken as 112 lpcd and for partially plumbed house is taken as 65 lpcd. Similarly, livestock demand is taken as 20% of the domestic demand and so on. As so many factors are involved in the demand of water, it is not possible to accurately determine the actual demand.

Some of the factors governing the water demand are described below:

i. Population:

Water demand increases with population size and population growth.
Future population projection is essential for designing water supply systems.

ii. Climate:

Hot and dry climates increase water consumption for drinking, bathing, and irrigation.
Cooler climates generally require less water.

iii. Standard of living:

Higher living standards lead to greater use of water through showers, washing machines, gardens, and household appliances.

iv. Quality and availability of water:

A reliable supply of safe drinking water encourages higher consumption.
Scarcity or poor quality of water reduces demand.

v. Metering of water supply:

Metered connections reduce wastage and promote efficient water use.
Unmetered supply often leads to excessive consumption.

c. Hydraulic grade line:

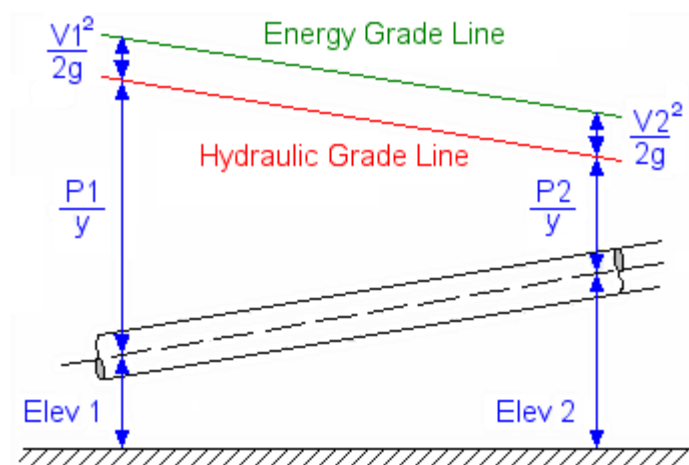
Hydraulic Grade Line (HGL) is an imaginary line joining the levels to which water would rise in piezometer tubes connected to a pipeline. It represents the sum of pressure head and elevation head of flowing water.

$$\text{HGL} = z + \frac{p}{\gamma}$$

where :

z = elevation head

$\frac{p}{\gamma}$ = pressure head



- HGL shows the variation of pressure along a pipeline.
- It always slopes downward in the direction of flow due to head loss from friction.
- In open channels, the HGL coincides with the water surface.
- In pressurized pipes, the HGL lies above the pipe centerline when the pressure is positive.

Use of HGL:

- Used in the design and analysis of water supply and pipe distribution systems.
- Helps determine the pressure available at different points in a pipeline.
- Identifies locations of low or negative pressure that may cause cavitation or air entry.
- Assists in selecting pump head and reservoir elevation.

3. **Calculate average water demand in litre per second (lps) for typical village with population of 1000. Find future average and design flow after 20 years. Define peak factor.**

Answer:

Present population (P_0) = 1000

Per capita water demand (q) = 45 LPCD

Population growth rate (r) = 2% per year

Design period = 20 years

Peak factor = 2.5

$$Q = \frac{P \times q}{86400} \quad Q = \frac{1000 \times 45}{86400} = 0.521 \text{ L/s}$$

Using geometric growth:

$$P_{20} = P_0(1+r)^n$$

$$= 1000(1.02)^{20}$$

$$= 1486$$

$$Q = \frac{1486 \times 45}{86400} = 0.774 \text{ L/s}$$

Future average water demand = 0.77 L/s

$$Q_d = \text{Peak Factor} \times \text{Average Flow}$$

$$= 2.5 \times 0.774 = 1.935 \text{ L/s}$$

Hence, the design flow = 1.94 L/s

The ratio of the maximum water demand to the average water demand is known as peak factor.

$$\text{Peak Factor} = \frac{\text{Maximum Demand}}{\text{Average Demand}}$$

It is used to account for short periods of high-water consumption so that water supply systems are designed to meet peak demand without interruption.

Typical values range from 2.0 to 3.0, with 2.5 commonly used for village water supply design.

4. **It is possible to have no reservoir for a gravity flow scheme? Why are elevated reservoirs provided in water supply systems? Explain briefly.**

Answer:

Yes, it is possible under certain conditions. A gravity flow scheme can operate without a reservoir if:

- The source provides a continuous and reliable discharge throughout the day.
- The source discharge is equal to or greater than the peak demand.
- The source is located at a higher elevation than the service area, providing sufficient pressure.
- Water demand does not fluctuate significantly.

However, in practice, a reservoir is generally provided because water demand varies during the day while the source flow is usually constant.

An elevated reservoir also known as overhead tank stores treated water at a higher elevation and supplies it to consumers by gravity. It is provided in the supply systems because of the advantages mentioned below:

- i. Elevated reservoir balances fluctuations between water supply and consumer demand.
- ii. It maintains adequate and uniform pressure in the distribution system.
- iii. It helps to store water for peak demand periods.
- iv. Elevated reservoir also provides emergency storage during power failure, pump breakdown, or source interruption.
- v. It ensures continuous water supply.
- vi. The reservoir also provides reserve water for fire-fighting purposes.
- vii. It helps to reduce the required pumping capacity by allowing pumps to operate at a more uniform rate.

5. **What are the main differences between basic principles of design of sewers and water mains? Explain the significance of minimum and maximum velocities of flow in the design of sewers.**

Answer:

A sewer is an underground conduit designed to convey wastewater and surface runoff to treatment facilities or disposal points at partial flow condition. Water mains on the other hand are the pipelines that are used to transport water from the source or reservoir to household under full flow conditions.

Sewers	Water mains
i. Sewers are designed to carry wastewater and sewage.	i. Water mains are designed to carry potable water.
ii. They are generally designed to flow partially full under gravitational force.	ii. They are designed to flow completely full under pressure.
iii. They are designed to achieve self-cleansing velocity to prevent sediment deposition.	iii. They are designed to maintain adequate pressure and acceptable flow velocity.
iv. Pipes are laid with uniform downward gradient.	iv. Pipes follow the ground profile and need not have continuous slope.
v. Leakage from sewers causes pollution of soil and groundwater.	v. Leakage causes water loss and pressure reduction.
vi. Ventilation is necessary to remove gases and odors.	vi. Ventilation is not required in water mains.
vii. Manholes are provided at the regular intervals for inspection and cleaning.	vii. Valves, hydrants and air valves are provided for operation and maintenance.

Self- cleansing velocity:

- It is the minimum velocity at which the solid particles remains in suspension without settling at sewer's bottom.
- It is the velocity at which even scour of the deposited particles of the given size will take place.
- It must be maintained at least once a day as it is not possible to maintain the velocity throughout the day.

Formula for Self-Cleansing Velocity

According to Shield's Theory:

$$V_s = \sqrt{\frac{8K(\rho_s - \rho)gd}{f\rho}}$$

Where:

V_s = Self-cleansing velocity (m/s)

K = Dimensionless constant

ρ_s = Density of solid particles (kg/m³)

ρ = Density of water (kg/m³)

g = Acceleration due to gravity (9.81 m/s²)

d = Diameter of particle (m)

f = Friction factor

Non-scouring velocity in the sewer:

- It is the maximum velocity at which interior sewer surface is not damaged.
- It is the velocity at which no scour or abrasion takes place.
- Non-Scouring Velocity prevents erosion of channel bed and banks and ensures stability of sewers.
- **Formula for Non-Scouring Velocity**

According to Kennedy's Theory:

$$V_0 = 0.55 m D^{0.64}$$

Where:

V_0 = Non-scouring (critical) velocity (m/s)

m = Critical velocity ratio

D = Depth of flow (m)

For average silt conditions:

$$m = 1.0$$

Therefore,

$$V_0 = 0.55 D^{0.64}$$

Significance of maximum and minimum velocity in the sewer:

- i. They prevent deposition of solids and blockages.
- ii. These velocities also help to avoid erosion and damage to sewer pipes.
- iii. They also ensure efficient transport of sewage.
- iv. It helps to reduce maintenance costs.
- v. Minimum and maximum velocities also increase the durability and service life of the sewer system.

6. Write down short notes on:

a. B.O.D

b. Ecological sanitation

c. Inverted siphon

Answer:

a. B.O.D.

Biochemical Oxygen Demand (BOD) is the amount of dissolved oxygen required by microorganisms to decompose biodegradable organic matter present in wastewater.

BOD treatment is necessary for the domestic waste before discharging in natural rivers because:

i. Prevents the depletion of natural resources:

Domestic sewage consists of large amount of organic matter. The discharge of untreated domestic sewage directly into river pollutes the natural resources. Microorganisms consumes the dissolved oxygen to decompose organic matter present in the sewage. This increases the BOD of sewage which rapidly decrease the DO content of river. Hence, to prevent depletion of natural resources, BOD treatment is necessary.

ii. Protects aquatic life:

Due to low DO level, aquatic life can be threatened as they require sufficient DO for their survival. Severe oxygen depletion results in destruction of aquatic ecosystems. Hence, domestic sewage should be adequately treated before disposing it to rivers.

iii. Maintains ecological balance:

High BOD leads to growth of micro-organisms and bacteria which disturbs the natural balance of ecosystem. Hence, disposal of domestic sewage only after treatment must be seriously considered.

iv. Prevents septic condition:

Excessive organic matter in sewage creates anaerobic conditions which gives rise to foul smell due to production of hydrogensulphide, methane and ammonia. This makes river water unpleasant and unsuitable for utilization. Hence, domestic sewage must not be directly disposed into river.

v. Prevents the spread of epidemics and diseases:

Polluted rivers may lead to spread of diseases and epidemics. When the polluted water from the river is used in any way by human or cattle, the pathogens and harmful contaminants present in the water might put serious effect on the health.

The necessity of BOD treatment can be observed in the Bagmati and Bishnumati rivers of Kathmandu Valley. The direct discharge of untreated domestic and industrial wastewater into these rivers has severely deteriorated their water quality. While the rivers are relatively clean upstream, they become turbid, polluted, and foul-smelling as they pass through Kathmandu. This highlights the importance of treating wastewater and reducing BOD before its discharge into natural water bodies. Delay in action might cost the loss of another natural resource and disbalance in the ecosystem.

b. Ecological sanitation:

Ecological Sanitation also known as EcoSan, is a sustainable sanitation approach that safely treats and reuses human excreta as a resource rather than treating it as waste. It aims to protect public health, conserve water, and recycle nutrients back to the soil. Ecological sanitation approach is suitable for rural communities, water-scarce areas, schools and institutions. According to UNICEF, 40 out of 100 people in Nepal have access to safely managed on-site sanitation and about 53% use safely managed sanitation services with roughly 62% to provide general improved facilities nationwide.

The examples of EcoSan are Urine Diverting Dry toilets (UDDT), Composting toilets, Double-vault Ecosan toilets. In Nepal, Ecological sanitation approach is actively promoted by organizations like the Environment and Public Health Organisation (ENPHO) and WaterAid in Nepal. EcoSan double-vault and urine-diverting toilets are mostly concentrated in agricultural peri-urban and rural clusters such as Siddhipur, Lubhu, and Kirtipur in the Kathmandu Valley.

Advantages:

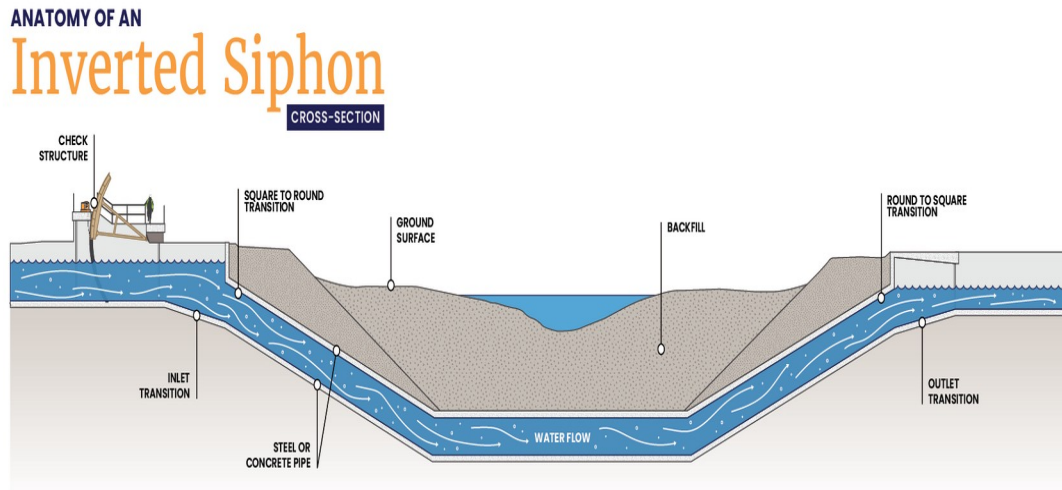
- Conserves water by reducing or eliminating flushing.
- Recycles nutrients such as nitrogen, phosphorus, potassium for agriculture.
- Reduces environmental pollution.
- Improves soil fertility and crop production.
- Suitable for water-scarce and rural areas.
- Low operation and maintenance costs.

Disadvantages:

- Requires user awareness and proper maintenance.
- Initial construction cost may be higher.
- Social and cultural acceptance may be low.
- Improper handling can pose health risks.

c. Inverted siphon:

An inverted siphon is a closed pressure conduit that transport fluids under an obstacle such as a river, railway, road, or valley when gravity flow is not possible. It flows under pressure due to the difference in hydraulic head. Inverted siphon is used while passing under deep valleys or other underground utilities. An inverted siphon works through gravity-driven flow, hydrostatic pressure, and the principle of communicating vessels rather than atmospheric vacuum.



Components of inverted siphon:

- Inlet chamber
- Pressure pipe (barrel)
- Outlet chamber
- Manholes for inspection and maintenance.

Advantages:

- Enables sewers to cross rivers, roads, and valleys.
- Maintains continuity of the sewer system.
- Suitable where gravity sewers cannot be laid directly.

Disadvantages:

- Higher construction and maintenance cost.
- Sediment deposition may occur if velocity is low.
- Difficult to inspect and clean.

7. What is self-cleansing velocity? How is it obtained in case of sewers?

Answer:

Self- cleansing velocity:

- It is the minimum velocity at which the solid particles remains in suspension without settling at sewer's bottom.
- It is the velocity at which even scour of the deposited particles of the given size will take place.
- It must be maintained at least once a day as it is not possible to maintain the velocity throughout the day.
- It is obtained by using shield's formula.

Formula for Self-Cleansing Velocity

According to Shield's Theory:

$$V_s = \sqrt{\frac{8 K (\rho_s - \rho) g d}{f \rho}}$$

Where:

V_s = Self-cleansing velocity (m/s)

K = Dimensionless constant

ρ_s = Density of solid particles (kg/m³)

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8. Describe the procedures laid down in the guidelines on the implementation and management of water supply project, 2047 for the formation of water user's committee. Also, mention its duties and responsibilities.

Answer:

- The guidelines on implementation and management of water supply project, 2047 (Procedural Guidelines for Water Supply Projects) lay down a stepwise, community-based procedure for forming a Water Users' Committee (WUC).
- It also defines its duties and responsibilities in planning, construction, operation, and maintenance of the scheme.
- WUC is a democratically formed local body that represents all users and is legally responsible for managing the project on behalf of the community.
- Procedures that are mentioned in the guidelines on the implementation and management of water supply project, 2047 for the formation of water user's committee are listed below:

i. Community mobilization and initial meetings:

- The implementing agency (DWSS/DWSSM, NGO, or project office) first conducts community consultation and awareness meetings in the service area.
- Objectives of the water supply project, community roles, and the need for a Water Users' Committee are explained.
- All the households/users are encouraged to participate, including women, disadvantaged and marginalized groups.

ii. Identification of users and service area:

- The guideline requires clear identification of beneficiaries and the geographical service area.
- Only the actual users of the proposed scheme are recognized as members of the "water users' group," who will elect the committee.

iii. Formation of user's group and discussion on committee structure:

- A general assembly of users is convened, where the community agrees on:
 - The size of the committee (usually 9–13 members, depending on scheme size).
 - Representation of different toles/wards and inclusion of women and disadvantaged groups.
 - Positions such as Chairperson, Vice-chairperson, Secretary, Treasurer and ordinary members.

iv. Election of committee members:

- The guidelines emphasize democratic selection normally through consensus or voting in the general assembly.
- Users nominate and elect individuals who are:

- Respected and trusted by the community.
 - Willing to work voluntarily for the project.
- Literate persons are preferred for secretary and Treasurer position.
- v. Documentation:
- Minutes of the general assembly, list of elected members, and their positions are recorded and signed by community representatives and project staff.
 - The Water Users' Committee is formally recognized by the project as the implementing partner, and its formation is endorsed by the local government body (VDC/municipality) as required.
- vi. Registration and legal recognition:
- As per associated legal provisions (Water Resources Act/Rules, Drinking Water Rules), the WUC may be **registered as a Water Users' Association** with the concerned authority (DCC, municipality, or Drinking Water Office).
 - Registration provides legal status to enter into agreements, operate bank accounts, and own/operate the scheme.

Duties and responsibilities of WUC:

- i. Participate in feasibility study, site selection and approve the final design and implementation plan.
- ii. Coordinate with local government (VDC/municipality) and other stakeholders to avoid conflicts over source use, land acquisition, and right-of-way.
- iii. Collect user fees and contributions regularly and deposit into a committee bank account to ensure transparent record keeping and no misuse of funds.
- iv. Organize and supervise community labor contributions (e.g., trench digging, material transport).
- v. Arrange routine inspection and maintenance of the system such as intake, pipelines, reservoir, tap stands, valves etc.
- vi. Hold regular committee meetings and inform the community about activities, income, expenditure, and major decisions.
- vii. Prepare periodic reports for the project office and local authority as required by the guideline.

9. Give the definition of environmental impact assessment (EIA). Explain some of the major government rules and regulations for EIA.

Answer:

- "Environmental impact assessment" is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- Projects requiring EIA are listed in **schedule 3 of EPR 2077**.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → EIA Report Preparation → Recommendation and submission → final review and approval → Decision making
- EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
- EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Irrigation project with command area greater than 2000 hectares.
 - c. Hydropower projects with installed capacity of greater than 50 MW.

Some of the major government rules and regulation for EIA is listed below in sequence:

i. **Screening:**

- This step determines whether the proposed project requires IEE or EIA.
- No project requiring EIA can be implemented without approval of the environmental study report.

ii. **Preparation of Scoping Document:**

The proponent prepares Scoping Document following the format in **Schedule-5** and is submitted to the concerned approving authority.

- For federal projects, it is submitted to the **Concerned Ministry** (e.g., the Ministry of Irrigation or Physical Infrastructure).
- The Concerned Ministry reviews it, adds its opinion, and forwards it to the **Ministry of Forests and Environment (MoFE)** within **15 days**.
- MoFE must approve or modify the Scoping Report within **15 days** of receiving the recommendation.

iii. **Preparation and Approval of work schedule:**

- The proponent prepares a **Work Schedule** (per **Schedule-8**) that outlines the methodology and timeline for the study.
- The Scoping Report and Work Schedule are often submitted and approved together.
- Once reviewed, MoFE must approve the Work Schedule within **15 days**.

iv. **Field study and Public Hearing:**

- The study must be carried out by a team of qualified experts who meet the academic and experience criteria listed in **Schedule-13**.
- The proponent **must organize at least one public hearing** in the project area to gather feedback from local communities, forest user groups, and local representatives.
- The proponent **must publicize the hearing** via local radio, newspapers, or other media, and post notices at the ward office. Documentation of the hearing (attendance, suggestions, photos, and audio-visual recordings) must be included in the final report.

v. **EIA Report Preparation:**

- EIA report is prepared based on the approved Scoping Report, Work Schedule, and the suggestions from the public hearing according to the detailed format in **Schedule-12**.
- **Public Notice for Feedback:** The proponent must issue another **7-day notice** (Schedule-9) in a national daily newspaper to gather written suggestions on the potential environmental impacts of the project.
- Reports are generally prepared in **Nepali** language.
- For foreign-invested projects, they may be in English, provided a concise **executive summary in Nepali** is included.

vi. **Recommendation and Submission:**

- The proponent must include a formal **Recommendation Letter (सिफारिस पत्र)** from the concerned **Local Level** and the relevant thematic office (per **Schedule-14**). These bodies must provide the recommendation within **15 days**.
- **Submission:** For federal projects, the report is submitted to the **Concerned Ministry**.
- The Concerned Ministry reviews the report and forwards it to **MoFE** with its formal opinion within **15 days**.
- The final report must be submitted within **two years** of the Scoping/Work Schedule approval.

vii. **Final Review and Approval of EIA:**

- **Final Public Review:** Upon receiving the EIA report, MoFE must make it public for **7 days** on its website and via a national daily newspaper to collect final public comments.
- **Review Committee:** MoFE may form a **committee** of representatives and experts to investigate the report, which may include on-site inspections or monitoring.
- **Approval Decision:** If the investigation shows the project will not cause significant adverse environmental impacts, MoFE shall approve the EIA report within **35 days**. The approval may include specific, binding conditions that the proponent must follow.
- Additionally, the project must start within **three years** of approval, or the proponent may be required to conduct a new study.

Note on Violations: Implementing a project without an approved EIA report or in violation of its conditions can lead to a fine of up to **5 million (50 लाख) rupees**.

10. **The role of government agencies has been transformed to the role of a facilitator from the provider of the system. How do you look at this statement in the context of sustainable development of water supply and sanitation sector?**

Answer:

In the past, government agencies were responsible for planning, financing, constructing, operating, and maintaining water supply and sanitation systems. Today, their role has shifted to that of a facilitator, while communities, local governments, NGOs, and the private sector take a more active role. This approach promotes sustainability, local ownership, long-term utilization of services and efficient service delivery.

Role of government as facilitator includes:

- Formulate policies, laws, and standards.
- Prepare plans and provide technical guidance.
- Mobilize financial resources and subsidies.
- Build the capacity of local governments and Water Users' Committees (WUCs).
- Monitor, regulate, and evaluate project performance.
- Ensure water quality, environmental protection, and public health.
- Coordinate among stakeholders and development partners.

As the role of government has changed into facilitator rather than provider, the benefits it has done for encouraging sustainable development are listed below:

- i. Encourages community participation and sense of ownership.
- ii. Enhances transparency and accountability during the construction process.
- iii. Ensures better cost recovery through user participation.
- iv. Improves operation and maintenance of water supply systems.
- v. Increases long-term sustainability of projects.
- vi. Promotes inclusive access to water and sanitation.